

$$\bullet \vec{r}_s = (0, -vt) = -vt \hat{e}_y \quad \times$$

$$\vec{r}_\Sigma = \xi \hat{e}_\xi + \eta \hat{e}_\eta$$

→ [143] → shared terms were zero, expanded

$$\textcircled{1} \quad \ddot{\vec{r}}_{\Sigma, \text{app}} = \omega^2 \vec{r}_\Sigma - 2\omega (\dot{\xi} \hat{e}_\eta - \dot{\eta} \hat{e}_\xi) \quad \times$$

$$\textcircled{2} \text{ Find path } \vec{r}_\Sigma = (\xi(t), \eta(t))$$

$$\hookrightarrow \text{(LHS)} \quad \ddot{\vec{r}}_{\Sigma, \text{app}} = \ddot{\xi}(t) \hat{e}_\xi + \ddot{\eta}(t) \hat{e}_\eta \quad (\text{Notes})$$

• expand RHS:

$$\begin{aligned} \text{(RHS)}: & \omega^2 (\xi \hat{e}_\xi + \eta \hat{e}_\eta) - 2\omega (\dot{\xi} \hat{e}_\eta - \dot{\eta} \hat{e}_\xi) \\ = & (\omega^2 \xi + 2\omega \dot{\eta}) \hat{e}_\xi + (\omega^2 \eta - 2\omega \dot{\xi}) \hat{e}_\eta \end{aligned}$$

$$\begin{aligned} \ddot{\xi}(t) &= \omega^2 \xi + 2\omega \dot{\eta} & \text{and } \leftarrow & \times \\ \ddot{\eta}(t) &= \omega^2 \eta - 2\omega \dot{\xi} & \leftarrow & \times \end{aligned}$$

$$\ast \text{ Hint: } \underline{z(t) \equiv \xi(t) + i\eta(t)} \quad \ast$$

$$\dot{z}(t) = \dot{\xi}(t) + i\dot{\eta}(t) \quad \ast$$

$$\underline{\ddot{z}(t)} = \ddot{\xi}(t) + i\ddot{\eta}(t)$$

$\ast$ Take $\ddot{\xi} + i\ddot{\eta}$ from above

$$\underline{\ddot{\xi} + i\ddot{\eta}} = (\omega^2 \xi + 2\omega \dot{\eta}) + i(\omega^2 \eta - 2\omega \dot{\xi}) = \underline{\ddot{z}}$$

• Collect ω terms

$$= \omega^2 (\xi + i\eta) + 2\omega (\dot{\eta} - i\dot{\xi})$$

$$= \omega^2 z - 2\omega \left(\frac{i}{i} \right) (\dot{\eta} - i\dot{\xi})$$

$$= \omega^2 z - 2 \frac{\omega}{i} (\dot{\tilde{z}} + i\dot{\tilde{r}})$$

$$\Rightarrow \left\{ \frac{1}{i} (\dot{\tilde{r}} - i\dot{\tilde{z}}) = \frac{1}{i} (i\dot{\tilde{r}} - i^2 \dot{\tilde{z}}) = \frac{1}{i} (\dot{\tilde{z}} + i\dot{\tilde{r}}) \right\}$$

$$\bullet \frac{1}{i} = -i$$

$$= \omega^2 z - 2\omega i \underbrace{(\dot{\tilde{z}} + i\dot{\tilde{r}})}_{\dot{\tilde{z}}}$$

$$\ddot{\tilde{z}} = \omega^2 z - 2\omega i \dot{\tilde{z}}$$

Solve this ODE:

$$\ddot{\tilde{z}} + 2\omega i \dot{\tilde{z}} - \omega^2 z = 0 \quad *$$

$$\bullet \frac{d^2 z}{dt^2} + 2\omega i \frac{1}{z} \frac{dz}{dt} - \omega^2 = 0$$

$$* \left(\frac{d}{dt} + i\omega \right) \underbrace{\left(\frac{d}{dt} + i\omega \right) z}_{u} = 0$$

$$\left(\frac{d}{dt} + i\omega \right) \underbrace{\left(\frac{dz}{dt} + i\omega z \right)}_u = 0$$

$$\bullet \left(\frac{d}{dt} + i\omega \right) u = 0$$

$$\frac{du}{dt} + i\omega u = 0 \Rightarrow \frac{du}{dt} = -i\omega u$$

$$\int \frac{du}{u} = \int -i\omega dt$$

$$\ln(u) = -i\omega t + C$$

$$u(t) = e^C \cdot e^{-i\omega t} \\ = A e^{-i\omega t}$$

$$A = e^C$$

$$u(t) = A e^{-i\omega t}$$

$$\hookrightarrow \left(\frac{dz}{dt} + i\omega z = A e^{-i\omega t} \right) e^{i\omega t}$$

$$\bullet \mu = e^{i\omega \int dt} = e^{i\omega t}$$

$$e^{i\omega t} \frac{dz}{dt} + i\omega e^{i\omega t} z = A$$

$$\int \frac{d}{dt} (e^{i\omega t} z) dt = \int A dt$$

$$e^{i\omega t} z = At + C$$

$$z = e^{-i\omega t} (At + C)$$

$$z = \xi(t) + i\eta(t)$$

$$e^{-i\omega t} = \cos(-\omega t) + i \sin(-\omega t)$$

$$= \cos(\omega t) - i \sin(\omega t)$$

$$z = (At + C) \cos(\omega t) - i (At + C) \sin(\omega t)$$

$$\underbrace{\hspace{10em}} \quad \underbrace{\hspace{10em}}$$

$$\xi(t)$$

$$\eta(t)$$

$$\xi(t) = (At + C) \cos(\omega t)$$

$$\eta(t) = -(At + C) \sin(\omega t)$$

$$\vec{r}_\Sigma = \xi(t) \hat{e}_\xi + \eta(t) \hat{e}_\eta$$

* Want A and C in terms of physical Quantities:

$$\vec{r}_S = (0, -Ut) = -Ut \hat{e}_y \quad \left| \quad \vec{r}_S = -U \hat{e}_y \right.$$

$$r_S(0) = (0, 0)$$

$$|\vec{r}_S| = U = V_S$$

$$\vec{r}_\Sigma(0) = (0, 0)$$

$$\text{ic) } \xi(0) = (A(0) + C) \cos(0) = \boxed{C = 0}$$

$$\hookrightarrow \xi(t) = At \cos(\omega t)$$

$$\eta(t) = -At \sin(\omega t)$$

$$\dot{\vec{r}}_{\Sigma, app} = \vec{v}_{\Sigma, app} \quad *$$

$$\textcircled{1} \dot{\xi}(t) = A \cos(\omega t) + A t [-\sin(\omega t)] \omega$$

$$\textcircled{2} \dot{\eta}(t) = -A \sin(\omega t) - A t \cos(\omega t) \omega$$

$$|\vec{v}_{\Sigma, app}(t)| \stackrel{!}{=} U \quad \leftarrow t=0$$

$$\vec{v}_{\Sigma, app}(t) = \dot{\xi}(t) \hat{e}_{\xi} + \dot{\eta}(t) \hat{e}_{\eta}$$

$$\vec{v}_{\Sigma, app}(0) = \dot{\xi}(0) \hat{e}_{\xi} + \dot{\eta}(0) \hat{e}_{\eta}$$

$$\textcircled{1} \dot{\xi}(0) = A$$

$$\dot{\eta}(0) = 0$$

$$\vec{v}_{\Sigma, app}(0) = A \hat{e}_{\xi}$$

$$|\vec{v}_{\Sigma, app}| = A$$

$$A = U \quad (\text{initial velocity})$$

$$\begin{aligned} \xi(t) &= U t \cos(\omega t) \\ \eta(t) &= -U t \sin(\omega t) \end{aligned}$$

* started with

$$\vec{r}_s(t) = -U t \hat{e}_y \quad \leftarrow \quad \vec{r}_s(0) = 0$$

$$\vec{v}_s(t) = \dot{\vec{r}}_s = -U \hat{e}_y$$

